Report: Experience, Data Language Requirements, and Salary Estimates in Data Analyst Job Postings

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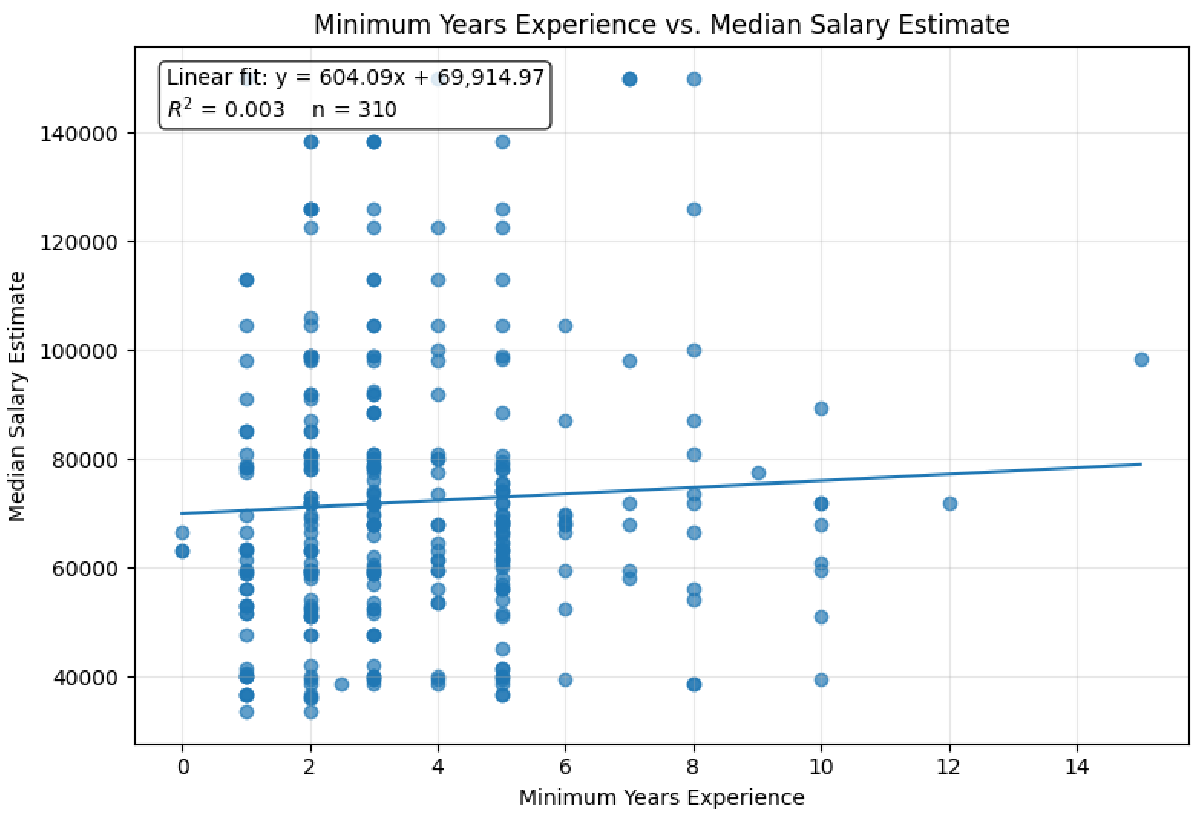
Brief introduction

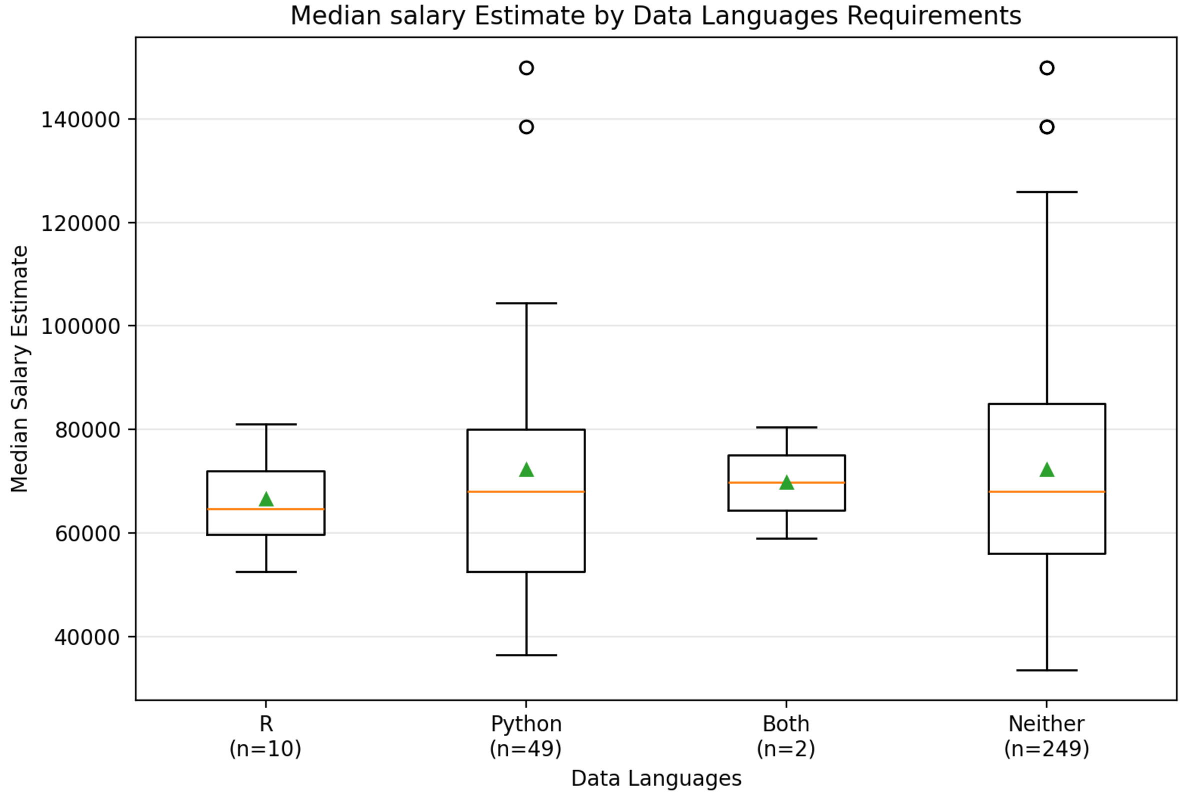
This dataset contains Data Analyst job postings with information on median salary estimates, minimum years of experience, and data language requirements. The aim of this project is to examine whether experience requirements and data language requirements are associated with differences in advertised salary estimates.

Analysis of years of experience



The scatterplot shows very little evidence of a meaningful linear relationship between minimum years of experience and median salary estimate. Although the regression line slopes slightly upward, the R^2 value is only 0.003, meaning that experience requirements explain almost none of the variation in salary estimates in this sample. This suggests that other factors, such as job title, location, company, industry, or seniority level, may be more important predictors of salary.

Analysis of median salary estimates by programming language requirements



The boxplot compares median salary estimates across data language requirement groups, but the group sizes are highly uneven. The Neither group contains most of the observations, while the R group is based on only 10 cases and the Both group on only 2 cases. Because of these small sample sizes, especially for R and Both, differences between groups should be interpreted cautiously and should not be treated as strong evidence of a broader pattern.

Overall conclusions

Overall, the figures suggest that neither minimum years of experience nor data language requirements show a strong association with median salary estimates in this dataset. However, the conclusions are limited by the small and uneven group sizes, suggesting the results should be viewed as exploratory rather than definitive.

Personal reflection on the process

This was again a surprising easy task. I learned that using tools like LLMs and APIs is more about integrating these tools into the individual steps of research instead of finding the perfect tool which can do it all. Now I just have to expand my toolbox as much as possible.